PRANVEER SINGH INSTITUTE OF TECHNOLOGY

Major Project Proposal

Team Id:

Team Details:

S No	Full Name	Roll No	Branch & Section	Mob No
1	Deepak Kanojiya	2301640100159	CS-3C	8318572901
2	Ashutosh Gandhi	2301640100126	CS-3C	7905195516
3	Athrva Prajapati	2301640100130	CS-3C	8299576191
4	Ayush Yadav	2301640100149	CS-3C	8604682099
5	Bhawani Shankar Upadhyay	2301640100154	CS-3C	7651908769
6	(Member)			

Project Title:

AtmosSense: A ML – Driven Weather Forecasting and Predictive Analysis Web Application

Domain: (Select/Tick all relevant Options)

1. **Software-Web Application**	2. Software-Mobile Application
3. **Artificial Intelligence/Machine Learning/Deep Learning**	4. Computer Vision/Image Processing
5. Blockchain	6. Internet of Things
7. Natural Language Processing	8. Big Data / Cloud Computing
9. Others (Specify if any) :	

Problem Statement:

Accurate and highly localized weather forecasting is critical for daily planning, agriculture, and disaster management. However, many conventional weather applications act merely as data aggregators, displaying third-party current conditions without offering customized predictive analytics. Furthermore, existing systems often fail to bridge the gap between live meteorological data and historical data trends. There is a need for an intelligent, scalable web platform that not only fetches real-time global weather metrics but also leverages historical datasets through machine learning to independently generate accurate, localized short-term forecasts (such as rain probability and hourly temperature trends) in an easily digestible, visually interactive format.

Proposed Solution:

The proposed solution is a dynamic, web-based weather forecasting platform built using the Python Django framework. The system integrates the OpenWeatherMap API to retrieve real-time meteorological data (temperature, humidity, wind speed, pressure, and visibility) for any user-queried city globally.

Beyond simply displaying live data, the backend utilizes scikit-learn to deploy pre-trained Machine Learning models based on historical weather datasets. It employs a Random Forest Classifier to predict the likelihood of rain the following day, and a Random Forest Regressor to forecast continuous variables like temperature and humidity for the upcoming 5-hour window. The frontend is designed with modern web technologies, featuring a responsive, dynamic UI with glassmorphism effects and interactive data visualization charts to display the predictive trends seamlessly to the end user.

Unique/Distinctive feature of the solution:

1. **Hybrid Real-Time & Predictive Engine:** Unlike standard apps that just display API data, this platform feeds live API metrics directly into a custom-trained Random Forest machine learning model to generate its own independent short-term forecasts.
2. **Dynamic Data Transformation:** Features a custom algorithmic pipeline that automatically converts continuous live data (e.g., numerical wind degrees from the API) into categorical features (compass directions) on the fly so it can be correctly processed by the ML model.
3. **Interactive Visualizations:** Implements an integrated Chart.js engine on the frontend that parses the backend Python predictions into interactive, gradient-styled line graphs for intuitive trend analysis.
4. **Context-Aware UI:** Utilizes dynamic background rendering that automatically updates the visual theme of the web application based on the real-time weather conditions (e.g., sunny, rainy, cloudy) of the searched location.

Tools/Technology Uses:

1) Hardware Requirements:
a) Processor: Intel i5 or higher
b) RAM: Minimum 8 GB
c) Storage: Minimum 256 GB SSD
d) Graphics Card: Not required (optional for UI enhancement)
e) Others: Stable internet connection

2) Software Requirements:

a) Operating System: Windows / Linux / macOS

b) Development Tools: VS Code, Python environment

c) Database: SQLite (Django default) or local CSV datasets for model training

d) Web Browser: Chrome / Edge / Firefox

e) Others:

- Django (Python Web Framework)
- Scikit-learn , Pandas , Numpy
- OpenWeatherMap API, pytz (Timezone management)
- HTML5, CSS3, Javascript
- Chart.js
- Bootstrap Icons

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(To be Filled by Faculty/Evaluator)

Proposal Evaluation:

1. Right Identification of the Problem (Appropriate selection of the problem)?
   a) Excellent   b) Good   c) Needs Improvement   d) Unacceptable

2. Relevance of the Solution (Adequately addressing the problem/need)?
   a) Excellent   b) Good   c) Needs Improvement   d) Unacceptable

3. Innovativeness in the Solution (Distinctive innovative components/features of the solution)?
   a) Excellent   b) Good   c) Needs Improvement   d) Unacceptable

4. Uniqueness of the Solution (Intellectual Property Component)?
   a) Excellent   b) Good   c) Needs Improvement   d) Unacceptable

Improvements/ Suggestions by the Evaluator:

1. 2. 3. 4.

Name of Faculty:

Designation:

Signature with Date:

Guidelines:

- One Proposal per team will be submitted by the team leader only.

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- A Team can have maximum 6 Members including team leader.
- Heading Font Size=14
- Content Font size =12
- Line Spacing: Single
- Font: Times New Roman
- Content Alignment: Justified

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Update department name in footer as per followings:

1. For CSE/CSE(Hindi) Core Students: Department of Computer Science and Engineering

For CSAI/AIDS Students: Department of Artificial Intelligence

For CSAIML/AIML Students: Department of Artificial Intelligence and Machine Learning

For DS/IOT/CYS Students Department of Data Science